

The free uniform spanning forest is a factor of IID

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Abstract

We prove that the free uniform spanning forest is a factor of IID on every infinite connected locally finite simple unweighted graph. One Borel rule works for all such graphs and uses no root, answering affirmatively the general factor question for unimodular random graphs. We also show that every translation-invariant strongly Rayleigh process indexed by a countable group is a factor of IID, including invariant determinantal processes with Hermitian positive-contraction kernels. This group-action conclusion requires no amenability.

1 Introduction

Uniform spanning forests are canonical probability measures on the edges of an infinite graph. Pemantle constructed infinite-volume spanning-tree limits on the integer lattice [16, Theorem 2.3]; Benjamini, Lyons, Peres and Schramm developed their general graph theory [3]. The free uniform spanning forest is obtained by taking weak limits of uniform spanning trees on finite connected exhaustions, without identifying boundary vertices. The limit exists and is independent of the exhaustion. We recall its classical construction from finite tree determinants in Proposition 2.2; see also Benjamini, Lyons, Peres, and Schramm [3, Section 5]. A factor of IID description asks whether this global random object can be constructed, equivariantly and measurably, from independent labels attached to the vertices. Lyons explicitly discussed the FUSF factor question for the group action on Cayley graphs in his report for the 2013 Oberwolfach workshop [12, p. 2392].

On transient graphs, the wired forest has an IID construction through cycle popping and Wilson’s algorithm rooted at infinity; see [3, Theorem 5.1]. Angel, Ray and Spinka give an explicit graph-factor formulation, retaining the transient graph scope and crediting the earlier construction [2, Section 4]. Determinantal structure also provides factor constructions in amenable group settings: Lyons and Thom prove Bernoulli isomorphism for equivariant determinantal measures on amenable Cayley graphs, with respect to the group action [13, Corollary 7.4]. This Bernoulli-isomorphism result concerns the specified group action; it does not by itself provide one rule on arbitrary graphs equivariant under every graph isomorphism.

For general graph laws, Timár proves FIID representability of the free forest on recurrent unimodular random graphs and, more generally, on invariantly amenable unimodular random graphs [17, arXiv version, Theorem 1 and Corollary 5]. Under invariant amenability he obtains the stronger conclusion of a finite-valued finitary factor [17, arXiv version, Theorem 2(1)]. His construction uses compatible monotone finite-volume limits along a hyperfinite exhaustion. The introduction to that December 2025 version states the unrestricted FUSF factor question as open. We prove the unrestricted graphwise statement by a different construction: the same Borel rule works for every graph in our class, independently of any random-graph law. The challenge is to obtain a single equivariant sample from the infinite-volume law while controlling all edges jointly.

1.1 Statement

The graphs in our main result are infinite, connected, locally finite, simple, and undirected. Write FUSF_G for the free uniform spanning forest law on $\{0, 1\}^{E(G)}$. Graphs with a distinguished undirected edge and vertex labels carry the usual Borel structure generated by finite marked neighborhoods. A rule is *root independent* if it uses no distinguished vertex. Equivariance means invariance under every isomorphism of labeled graphs carrying the distinguished edge to the distinguished edge.

Theorem 1.1. *There is one Borel, root-independent, equivariant rule*

$$\Phi(G, U, e) \in \{0, 1\}$$

such that, for every infinite connected locally finite simple undirected unweighted graph G and independent $\text{Uniform}[0, 1]$ vertex labels $U = (U_v)_{v \in V(G)}$, the entire edge set

$$\{e \in E(G) : \Phi(G, U, e) = 1\}$$

has law FUSF_G .

Corollary 1.2. *The free uniform spanning forest is a graph factor of IID under every unimodular law supported on the stated class of graphs. The same rule works for every such law.*

Here unimodularity means the mass-transport identity

$$\mathbb{E} \sum_{v \in V(G)} f(G, o, v) = \mathbb{E} \sum_{v \in V(G)} f(G, v, o)$$

for every nonnegative Borel function of a graph with two distinguished vertices. Theorem 1.1 is a graphwise statement and does not use this identity. In particular, no amenability, transience, uniform degree bound, or degree moment is assumed. The result asserts an ordinary Borel factor, without a finitary coding radius or a finite input alphabet.

1.2 Strongly Rayleigh processes

The sampling criterion also applies to a broader class of binary laws. For a finite set F , a probability measure μ on $\{0, 1\}^F$ is *strongly Rayleigh* if its generating polynomial

$$g_\mu(z) = \sum_{x \in \{0, 1\}^F} \mu(x) \prod_{i \in F} z_i^{x_i}$$

is nonzero whenever $\text{Im } z_i > 0$ for every $i \in F$. For a countable index set, we require every finite-coordinate marginal to be strongly Rayleigh, following Borcea, Brändén, and Liggett [4, Definition 2.13].

For a countable group Γ , write τ_g for the regular left action on coordinate families, given by $(\tau_g a)_h = a_{g^{-1}h}$.

Theorem 1.3 (Strongly Rayleigh processes on countable groups). *Let Γ be a countable group and let ν be a strongly Rayleigh probability law on $\{0, 1\}^\Gamma$ invariant under every τ_g . There is a Borel map*

$$\Phi_\nu : [0, 1]^\Gamma \longrightarrow \{0, 1\}^\Gamma$$

such that $\Phi_\nu(\tau_g u) = \tau_g \Phi_\nu(u)$ for every $g \in \Gamma$ and every $u \in [0, 1]^\Gamma$, and such that $\Phi_\nu(U)$ has law ν when the coordinates of U are independent $\text{Uniform}[0, 1]$ variables.

For a self-adjoint positive contraction K on $\ell^2(\Gamma)$, the determinantal law of a random subset $X \subseteq \Gamma$ is characterized by the finite inclusion probabilities

$$\mathbb{P}(A \subseteq X) = \det(K_{g,h})_{g,h \in A} \quad \text{for every finite } A \subseteq \Gamma.$$

Corollary 1.4 (Determinantal processes on countable groups). *Let Γ be a countable group and let K be a self-adjoint positive contraction on $\ell^2(\Gamma)$, so $0 \leq K \leq I$. If the determinantal probability law with kernel K is invariant under the regular left action, then it is a factor of IID under that action. In particular, this holds whenever K commutes with the left regular representation of Γ .*

These statements concern the regular action on the group itself, not an arbitrary action on another index set. The factor may depend on Γ and ν . No amenability or finite-generation assumption is imposed, and finite groups are allowed. The conclusion is a Borel factor of IID; it does not assert a finitary coding or a Bernoulli isomorphism. In particular, it is distinct from the amenable-group Bernoulli-isomorphism result of Lyons and Thom cited above. Their broader determinantal factor question appears in [13, Question 7.7]; Corollary 1.4 treats the regular action on the group itself.

1.3 The construction

Let X have the target forest law and observe its edge indicators through independent Brownian noises:

$$Y_e(t) = tX_e + B_e(t).$$

For finitely many observed edges, the posterior is an exponential tilt of the forest law. The main quantitative input is the following response identity for a finite weighted spanning tree. If p_e is the probability of including edge e and h_f is the logarithm of the weight of edge f , then

$$\sum_f \left| \frac{\partial p_e}{\partial h_f} \right| = 2p_e(1 - p_e) \leq \frac{1}{2}.$$

Negative off-diagonal covariances and the fixed size of a finite tree give this identity. Passing it through finite-volume limits yields an everywhere-defined posterior drift $b(t, y)$ whose response to bounded changes of the field is uniformly Lipschitz, even when the fields themselves are unbounded across the graph.

Subtracting the posterior drift gives the innovations

$$\widehat{W}_e(t) = Y_e(t) - \int_0^t b_e(s, Y(s)) ds.$$

They are independent Brownian motions. The response bound makes the integral equation

$$Z_e(t) = w_e(t) + \int_0^t b_e(s, Z(s)) ds$$

uniquely solvable by a Borel Picard iteration for every continuous input field w . Only differences between iterates are bounded; there is no assumption that a countable Brownian field takes values in ℓ^∞ . Thus $Y = Z(\widehat{W})$, and independent Brownian inputs to this deterministic decoder reproduce the observation law. Integer-time slopes recover X .

Proposition 4.2 isolates the resulting sampling criterion for countable binary laws with uniformly controlled finite-tilt responses. The criterion separates the probabilistic observation argument

from the quantitative property of the target measure. For spanning forests, intrinsic edge-centered balls make the decoder equivariant across graphs. Finally, independent vertex labels supply the independent edge paths by a local allocation of disjoint random coordinates.

This construction adapts an established observation mechanism. Stochastic localization was introduced by Eldan [7]; a finite-dimensional form with fixed noise covariance appears in [8, Section 2, Proposition 7]. El Alaoui and Montanari [6, Section 3, Theorem 2] develop its Gaussian-channel observation and innovation formulation. A particularly close antecedent is Nam, Sly and Zhang [14, Theorem 1 and Section 2.2]: for free Ising measures on regular trees in their parameter range, they obtain FIID codings through Brownian-driven systems with posterior-mean drift and recover spins at large times. Here we prove the countable-coordinate posterior and joint innovation identities, then use the uniform response bound to obtain an inverse defined at every continuous input.

A companion [15, Theorem 1.1] gives the sharp FIID threshold for the free zero-field ferromagnetic Ising law on regular trees, including the critical case, with a factor that can be chosen to commute with every tree automorphism on every label input. Its conditional-copy recovery differs from our uniform-response inverse.

For strongly Rayleigh laws, symmetric homogenization supplies a fixed-size strongly Rayleigh extension, and the Rayleigh inequality gives nonpositive off-diagonal covariances [4, Definition 2.12 and Theorems 4.1–4.2]. The covariance-row bound then holds with the same constant $1/2$, as an inequality rather than the weighted-tree identity. Translates of one finite exhaustion of Γ make the resulting drift equivariant for the regular group action.

Section 2 reconstructs the free limit and establishes the response estimate and the canonical drift. Section 3 proves the posterior and innovation identities. The deterministic inverse and sampling criterion follow in Section 4. Section 5 constructs the single Borel graph rule and proves Theorem 1.1. Section 6 proves Theorem 1.3 and Corollary 1.4.

2 Weighted trees and a canonical response field

The analytic input is a response estimate whose constant is independent of the number of edges and of their weights. We first prove it on finite graphs, then use the free spanning-tree limit to construct a drift on every field of real edge marks. The finite determinant and transfer-current identities are classical; see Burton and Pemantle [5] and the weighted formulation in [3, Section 4]. We prove the needed identities and their free-limit consequence below.

Lemma 2.1 (Weighted-tree response). *Let H be a finite connected graph with at least two vertices, and let $\mathcal{T}(H)$ denote its set of spanning trees. For $h \in \mathbb{R}^{E(H)}$, give each $T \in \mathcal{T}(H)$ probability proportional to $\exp(\sum_{f \in T} h_f)$, and write $\xi_e = \mathbf{1}_{\{e \in T\}}$ and $p_e(h) = \mathbb{E}_h \xi_e$. For every $e \in E(H)$,*

$$\frac{\partial p_e}{\partial h_f} = \text{Cov}_h(\xi_e, \xi_f), \quad \sum_{f \in E(H)} \left| \frac{\partial p_e}{\partial h_f} \right| = 2p_e(h)(1 - p_e(h)) \leq \frac{1}{2}. \quad (2.1)$$

In particular, for all $h, h' \in \mathbb{R}^{E(H)}$,

$$|p_e(h) - p_e(h')| \leq \frac{1}{2} \max_{f \in E(H)} |h_f - h'_f|. \quad (2.2)$$

Proof. Set $c_f = e^{h_f}$. Orient the edges temporarily, delete one row of the oriented incidence matrix, and denote the resulting columns by a_f . The reduced weighted Laplacian is

$$L = \sum_{f \in E(H)} c_f a_f a_f^T.$$

Write $r = |V(H)| - 1$. Cauchy–Binet, applied to the reduced incidence matrix with column f multiplied by $\sqrt{c_f}$, gives

$$\det L = \sum_{\substack{F \subseteq E(H) \\ |F|=r}} \det(a_f : f \in F)^2 \prod_{f \in F} c_f = \sum_{T \in \mathcal{T}(H)} \prod_{f \in T} c_f =: \mathcal{Z}_H(h).$$

For the second equality, a set containing a cycle has dependent incidence columns, while an acyclic set of r edges is a spanning tree. Its reduced incidence determinant has absolute value one: expand at a leaf other than the vertex whose row was deleted and continue by induction. Connectedness supplies a spanning tree, so $\mathcal{Z}_H(h) > 0$ and L is invertible.

Differentiating the finite partition sum gives both $p_e = \partial_{h_e} \log \mathcal{Z}_H$ and the covariance identity in (2.1). Differentiating the determinant instead gives

$$p_e = c_e a_e^\top L^{-1} a_e.$$

Since $\partial_{h_f} L^{-1} = -c_f L^{-1} a_f a_f^\top L^{-1}$, for distinct edges e, f we obtain

$$\text{Cov}_h(\xi_e, \xi_f) = -c_e c_f (a_e^\top L^{-1} a_f)^2 \leq 0. \quad (2.3)$$

Every spanning tree has exactly r edges. Consequently

$$\sum_{f \in E(H)} \text{Cov}_h(\xi_e, \xi_f) = \text{Cov}_h\left(\xi_e, \sum_{f \in E(H)} \xi_f\right) = 0.$$

The diagonal covariance is $p_e(1 - p_e)$; all the other covariances are nonpositive by (2.3). Their absolute sum is therefore $2p_e(1 - p_e) \leq 1/2$, as claimed. This also covers a bridge: its inclusion probability is one and its entire covariance row is zero. Finally, integrate the gradient along the segment from h to h' to obtain (2.2). The estimate holds at every point of that segment, without a restriction on the weights. $\square$

2.1 The free limit from finite determinants

We include the free-limit construction, so that the passage from finite weighted trees to the infinite law has an explicit foundation. The relevant space is the closure of the finite cycle flows; no description of all square-summable divergence-free flows on the infinite graph is needed. This cycle-space description is classical; see [3, Proposition 7.1 and Theorem 7.8].

Proposition 2.2 (Free exhaustion limit). *Let G be an infinite connected locally finite simple graph. If $H_1 \subseteq H_2 \subseteq \dots$ are finite connected subgraphs with $\bigcup_n E(H_n) = E(G)$, the uniform spanning-tree laws on H_n , extended by absent edges outside H_n , converge weakly on $\{0, 1\}^{E(G)}$. Their limit is independent of the exhaustion, is supported on spanning forests, and is transported by every graph isomorphism. We denote this law by FUSF_G .*

Proof. Temporarily orient all edges of G . For a finite connected subgraph H with at least two vertices, let A_H be its reduced incidence matrix and put

$$K_H = A_H^\top (A_H A_H^\top)^{-1} A_H.$$

For each finite $F \subseteq E(H)$, the matrix-tree calculation in Lemma 2.1, followed by the rectangular determinant identity, gives the polynomial identity

$$\begin{aligned} \mathbb{E} \prod_{e \in F} (1 + z_e \xi_e) &= \frac{\det(A_H A_H^\top + A_F \operatorname{diag}(z) A_F^\top)}{\det(A_H A_H^\top)} \\ &= \det(I_F + \operatorname{diag}(z)(K_H)_{F,F}). \end{aligned} \quad (2.4)$$

Here A_F consists of the columns indexed by F . The first equality can first be read for $z_e > -1$ as a change of edge weights, and then holds for all z because both sides are polynomials. The identity $\det(I + UV) = \det(I + VU)$ used in the second line follows by eliminating blocks in $\begin{pmatrix} I & U \\ -V & I \end{pmatrix}$ in either order. Expanding the last determinant in its diagonal variables and comparing coefficients yields

$$\mathbb{P}(F \subseteq T_H) = \det(K_H)_{F,F}. \quad (2.5)$$

This includes sets larger than $|V(H)| - 1$: the corresponding minor and tree-inclusion probability are both zero. No minor is inverted.

Regard finite edge vectors as vectors of $\ell^2(E(G))$ by zero extension. Let $C_H = \ker A_H$ in the coordinates of $E(H)$, and let Q_H be the orthogonal projection onto those coordinates. The matrix K_H is the orthogonal projection onto the row space of A_H , so its zero extension is

$$K_H = Q_H - P_{C_H}. \quad (2.6)$$

Deleting a different incidence row does not change C_H : the full incidence matrix has entries summing to zero in each column, so zero divergence at all but one vertex forces zero divergence at the remaining vertex. The space C_H is exactly the span of its oriented cycle flows. Indeed, choose a spanning tree of H and subtract the fundamental cycle flow for each non-tree edge to eliminate that edge's coefficient. The remaining circulation is supported on a tree and vanishes by successive leaf removal.

Write $C_n = C_{H_n}$. These spaces increase under zero extension: a circulation supported on H_n still has zero divergence at every vertex of H_{n+1} . Every finite cycle of G is eventually contained in H_n , because the exhaustion includes every edge. Therefore the closure

$$C = \overline{\bigcup_n C_n}$$

is the closed span of all finite cycle flows of G , independent of the chosen exhaustion, whether or not its members are induced subgraphs. The projections P_{C_n} converge strongly to P_C . To see this directly, for $m \geq n$ and $x \in \ell^2(E(G))$ orthogonality gives

$$\|P_{C_m} x - P_{C_n} x\|^2 = \|P_{C_m} x\|^2 - \|P_{C_n} x\|^2.$$

The squared norms increase and are bounded by $\|x\|^2$, so the projected vectors are Cauchy. Their limit lies in C , and its difference from x is orthogonal to every C_n , hence to C . This identifies the limit as $P_C x$. Also $Q_{H_n} \rightarrow I$ strongly. Thus the zero-extended K_{H_n} converge strongly to $K = I - P_C$, and every fixed finite principal minor converges to that of K .

For disjoint finite edge sets A, B , inclusion-exclusion now gives a limit for

$$\mathbb{P}(A \subseteq T_{H_n}, B \cap T_{H_n} = \emptyset) = \sum_{J \subseteq B} (-1)^{|J|} \det(K_{H_n})_{A \cup J, A \cup J}$$

once $A \cup B \subseteq E(H_n)$; the empty determinant is one. These limits are nonnegative, normalized and consistent on every finite cube, since they are limits of actual probabilities. To construct

their countable law explicitly, temporarily enumerate the edges and recursively split $[0, 1)$ into half-open intervals of the prescribed binary-prefix masses. A uniform point selects one successive prefix at every depth and hence a configuration with these finite marginals. Zero-mass intervals may be omitted. Cylinder probabilities uniquely determine the resulting law, so the temporary enumeration affects neither the law nor the graph rule constructed later. Continuous functions on the compact countable binary product are uniformly approximable by finite-coordinate functions; cylinder convergence consequently proves weak convergence.

For every finite cycle, the event that all its edges are present has probability zero in every sufficiently late tree law and hence in the limit. There are only countably many finite cycles, so the limit is acyclic on the full vertex set. Reversing any orientations conjugates each finite principal matrix by a diagonal matrix with entries ± 1 and preserves its determinant. Graph isomorphisms transport the finite cycle space and all the limiting cylinder probabilities. This proves the claimed independence and invariance. $\square$

2.2 Finite tilts and the canonical drift

Fix now an infinite connected locally finite simple undirected graph G . Its edge set $E = E(G)$ is countable. Write $\nu = \text{FUSF}_G$ for the free uniform spanning forest law, viewed as a law on $\{0, 1\}^E$. For $e \in E$, finite $S \subseteq E$, $t \geq 0$, and $y \in \mathbb{R}^E$, define

$$b_{e,S}(t, y) = \frac{\int x_e \exp\left(\sum_{f \in S} (y_f - t/2)x_f\right) d\nu(x)}{\int \exp\left(\sum_{f \in S} (y_f - t/2)x_f\right) d\nu(x)}. \quad (2.7)$$

The denominator is positive, since its integrand is at least $\exp(\sum_{f \in S} \min\{y_f - t/2, 0\})$.

Lemma 2.3 (Finite-field forest response). *The function $b_{e,S}$ takes values in $[0, 1]$ and is continuous in $(t, (y_f)_{f \in S})$. For all $t \geq 0$ and $y, y' \in \mathbb{R}^E$,*

$$|b_{e,S}(t, y) - b_{e,S}(t, y')| \leq \frac{1}{2} \max_{f \in S} |y_f - y'_f|, \quad (2.8)$$

where the maximum over the empty set is defined to be zero.

Proof. Both integrals in (2.7) are finite sums over the possible configurations on $S \cup \{e\}$. Positivity of the denominator proves continuity, and the range follows from $0 \leq x_e \leq 1$. If S is empty, the function is the constant $\nu(x_e = 1)$.

For nonempty S , take a finite connected free exhaustion H_m of G . For all sufficiently large m , it contains $S \cup \{e\}$. Give its edges log weights $y_f - t/2$ for $f \in S$ and zero elsewhere. The resulting tree inclusion probability is the ratio in (2.7) with the uniform spanning-tree law on H_m in place of ν . Lemma 2.1 bounds the difference between this probability and the one using y' by the right side of (2.8). With S, t, y fixed, both integrands are bounded cylinder functions, so their expectations converge under the free spanning-tree limit. The limiting denominator is strictly positive by the displayed lower bound. Thus the ratios converge to $b_{e,S}(t, y)$, and likewise for y' . Passing to the limit proves (2.8). $\square$

To specify one drift at every field, we now fix the outer exhaustion intrinsically. If $e = \{u, v\}$, let $H_n(e)$ be the induced subgraph on

$$\{z \in V(G) : \min\{d_G(z, u), d_G(z, v)\} \leq n\}, \quad n \geq 1,$$

and set $S_n(e) = E(H_n(e))$. These graphs are finite and connected, increase with n , and exhaust G . Define, for every $t \geq 0$ and every $y \in \mathbb{R}^E$,

$$b_e(t, y) = \limsup_{n \rightarrow \infty} b_{e, S_n(e)}(t, y), \quad b(t, y) = (b_e(t, y))_{e \in E}. \quad (2.9)$$

Proposition 2.4 (Canonical drift). *Each b_e is a jointly Borel function on $[0, \infty) \times \mathbb{R}^E$ with values in $[0, 1]$, where $\mathbb{R}^E$ has its product Borel structure. The construction commutes with graph isomorphisms and does not select a root. For every $t \geq 0$ and $y, y' \in \mathbb{R}^E$ such that $\sup_{f \in E} |y_f - y'_f| < \infty$, it satisfies*

$$\sup_{e \in E} |b_e(t, y) - b_e(t, y')| \leq \frac{1}{2} \sup_{f \in E} |y_f - y'_f|. \quad (2.10)$$

Neither y nor y' is required to have bounded coordinates.

Proof. The range and Borel assertions follow from Lemma 2.3 and the countable limsup in (2.9). If $d = \sup_f |y_f - y'_f| < \infty$, that lemma gives

$$b_{e, S_n(e)}(t, y) \leq b_{e, S_n(e)}(t, y') + d/2$$

for every e, n . Taking limsups and then interchanging y, y' proves (2.10).

For the intrinsic nature of the construction, the inner limit has the explicit expression

$$b_{e, S_n(e)}(t, y) = \lim_{m \rightarrow \infty, m \geq n} \frac{\sum_{T \in \mathcal{T}(H_m(e))} \mathbf{1}_{\{e \in T\}} \exp \left(\sum_{f \in T \cap S_n(e)} (y_f - t/2) \right)}{\sum_{T \in \mathcal{T}(H_m(e))} \exp \left(\sum_{f \in T \cap S_n(e)} (y_f - t/2) \right)}. \quad (2.11)$$

This is exactly the fixed-finite-field limit used in the proof of Lemma 2.3. Each finite ratio is invariant under isomorphisms preserving the distinguished undirected edge and its field marks. The prescribed limits preserve that property. The same finite formulas therefore specify the drift as the graph varies; their use in a Borel graph rule will be made explicit below. $\square$

The order of the two spatial limits matters: in (2.11), n is fixed while m tends to infinity, and only afterwards is the limsup in n taken. We neither differentiate an infinite-volume partition function nor use a cardinality identity for an infinite forest. We also do not require the outer limit to exist at arbitrary fields. The limsup supplies a bounded Borel version everywhere, and (2.10) controls its change under every bounded perturbation of a possibly unbounded field.

3 Brownian observations and their innovations

Subtracting the posterior mean drift from the observations will produce a family of independent Brownian paths. Section 4 will use a uniform response bound to reconstruct the observations deterministically from this family.

The finite-dimensional innovation identity is classical in nonlinear filtering; see Fujisaki, Kallianpur, and Kunita [9, Lemma 2.2]. Here the probabilistic argument applies to any probability law ν on $\{0, 1\}^I$, where I is countable. For each $i \in I$, fix finite sets $S_n(i)$ increasing to I . Define $b_{i, S}$ by the finite-tilt formula (2.7), with this law ν , and define b_i by (2.9). These functions are bounded and jointly Borel. No response bound is needed in this section.

On a reference probability space let $X \sim \nu$, and independently let $(B_i)_{i \in I}$ be independent standard Brownian motions. Set

$$Y_i(t) = tX_i + B_i(t), \quad \mathcal{H}_t = \sigma\{Y_i(s) : i \in I, 0 \leq s \leq t\}. \quad (3.1)$$

We use these raw observation sigma-fields; no completion or right-continuity assumption on the filtration is necessary.

Lemma 3.1 (Posterior drift). *For every fixed $t \geq 0$ and $i \in I$,*

$$b_i(t, Y(t)) = \mathbb{E}[X_i \mid \mathcal{H}_t] \quad \text{almost surely.} \quad (3.2)$$

Proof. For finite $S \subset I$, write $\mathcal{H}_t^S = \sigma\{Y_j(s) : j \in S, 0 \leq s \leq t\}$. When $t > 0$, the density of the endpoint vector $(Y_j(t))_{j \in S}$, conditional on X , relative to independent $N(0, t)$ variables is

$$\exp\left(\sum_{j \in S} (y_j - t/2)X_j\right).$$

Here $X_j^2 = X_j$. Moreover,

$$Y_j(s) - \frac{s}{t}Y_j(t) = B_j(s) - \frac{s}{t}B_j(t), \quad 0 \leq s \leq t.$$

These bridge paths are jointly independent of X and the endpoint vector. Indeed, at finitely many times this follows from zero Gaussian covariances and independence of B from X ; continuity and rational-time evaluation extend it to the paths. Bayes' formula therefore gives

$$b_{i,S}(t, Y(t)) = \mathbb{E}[X_i \mid \mathcal{H}_t^S] \quad \text{almost surely.}$$

At $t = 0$ both sides equal $\mathbb{E}X_i$.

For fixed i, t , the sigma-fields $\mathcal{H}_t^{S_n(i)}$ increase to generate $\mathcal{H}_t$. The bounded conditional expectations $M_n = \mathbb{E}[X_i \mid \mathcal{H}_t^{S_n(i)}]$ converge almost surely to $\mathbb{E}[X_i \mid \mathcal{H}_t]$. We recall a short verification of this last step. For $m \geq n$,

$$\mathbb{E}|M_m - M_n|^2 = \mathbb{E}M_m^2 - \mathbb{E}M_n^2.$$

Choose $n_k \uparrow \infty$ so that this difference is at most 2^{-4k} whenever $m \geq n = n_k$. The maximal inequality obtained by conditioning on the first crossing gives

$$\mathbb{P}\left(\max_{n_k \leq r \leq m} |M_r - M_{n_k}| > 2^{-k}\right) \leq 2^{2k} \mathbb{E}|M_m - M_{n_k}|^2 \leq 2^{-2k}.$$

To see the first inequality, split into the disjoint events of a first crossing at r and apply conditional Jensen to $M_m - M_{n_k}$ at that stage. Letting $m \rightarrow \infty$ and summing over k proves that the full sequence is almost surely Cauchy. Its bounded limit has the same integral as X_i against every event in each finite-stage sigma-field. A monotone-class argument identifies it with $\mathbb{E}[X_i \mid \mathcal{H}_t]$. Thus the limsup defining b_i is an actual limit along the observations at this fixed time, and (3.2) follows. $\square$

Define the *innovations* by ordinary scalar time integrals:

$$\widehat{W}_i(t) = Y_i(t) - \int_0^t b_i(s, Y(s)) ds. \quad (3.3)$$

Lemma 3.2 (Joint innovation law). *The family $(\widehat{W}_i)_{i \in I}$ has the law of independent standard Brownian motions. Its finite-coordinate increments after time u are independent of $\mathcal{H}_u$.*

Proof. Fix $T < \infty$. The map $(s, \omega) \mapsto Y(s, \omega)$ on $[0, T]$ is measurable for $\mathcal{B}([0, T]) \otimes \mathcal{H}_T$, coordinate by coordinate. Composing with the bounded Borel drift shows that the integral in (3.3) is defined, measurable from the observation history, and continuous in its upper endpoint. Hence $\widehat{W}$ is coordinatewise continuous and adapted.

For $0 \leq u < v$ and any bounded $\mathcal{H}_u$ -measurable random variable K , Lemma 3.1 gives, separately for each $s \in [u, v]$,

$$\mathbb{E}[Kb_i(s, Y(s))] = \mathbb{E}[KX_i].$$

Bounded Fubini integrates these numerical identities. Since $B_i(v) - B_i(u)$ is independent of X and the Brownian histories through u , it is independent of $\mathcal{H}_u$. Consequently

$$\mathbb{E}[\widehat{W}_i(v) - \widehat{W}_i(u) \mid \mathcal{H}_u] = 0. \quad (3.4)$$

This argument uses no single exceptional set valid for every real time.

We identify the joint law using conditional characteristic functions: the error on each small interval is of order $h^{3/2}$, so the accumulated error vanishes under subdivision. Let $\theta = (\theta_i)_{i \in I} \in \mathbb{R}^I$ have finite support, and put

$$L_\theta = \sum_i |\theta_i|, \quad \sigma_\theta^2 = \sum_i \theta_i^2, \quad h = v - u.$$

The increment

$$D_{u,v} = \sum_i \theta_i (\widehat{W}_i(v) - \widehat{W}_i(u)) = Q + R$$

has a conditionally Gaussian term $Q = \sum_i \theta_i (B_i(v) - B_i(u))$ of conditional variance $h\sigma_\theta^2$, and a correction satisfying $|R| \leq hL_\theta$. It follows, without assuming that Q and R are independent, that

$$\begin{aligned} \left| \mathbb{E}[D_{u,v}^2 \mid \mathcal{H}_u] - h\sigma_\theta^2 \right| &\leq 2L_\theta\sigma_\theta h^{3/2} + L_\theta^2 h^2, \\ \mathbb{E}[|D_{u,v}|^3 \mid \mathcal{H}_u] &\leq 4(c_3\sigma_\theta^3 h^{3/2} + L_\theta^3 h^3), \end{aligned}$$

where $c_3 = \mathbb{E}|N(0, 1)|^3$. Together with (3.4), the Taylor bound $|e^{ix} - 1 - ix + x^2/2| \leq |x|^3/6$ yields a deterministic constant C_θ such that, for $h \leq 1$,

$$\left| \mathbb{E}[e^{iD_{u,v}} \mid \mathcal{H}_u] - e^{-h\sigma_\theta^2/2} \right| \leq C_\theta h^{3/2}. \quad (3.5)$$

Fix $s < t$ and $A \in \mathcal{H}_s$. Partition $[s, t]$ into N equal intervals with endpoints $t_0 = s, \dots, t_N = t$ and length $h \leq 1$. Set

$$a = e^{-h\sigma_\theta^2/2}, \quad \mu_k = \mathbb{E} \left[\mathbf{1}_A \prod_{r=1}^k e^{iD_{t_{r-1}, t_r}} \right].$$

The preceding factors are $\mathcal{H}_{t_{k-1}}$ -measurable and have modulus one. Thus (3.5) gives $\mu_k = a\mu_{k-1} + \delta_k$ with $|\delta_k| \leq C_\theta h^{3/2}$. Iterating, and using $0 < a \leq 1$, gives

$$\left| \mathbb{E}[\mathbf{1}_A e^{iD_{s,t}}] - \mathbb{P}(A) e^{-(t-s)\sigma_\theta^2/2} \right| \leq NC_\theta h^{3/2} \longrightarrow 0.$$

For every finite set of coordinates and every $A \in \mathcal{H}_s$, uniqueness of finite-dimensional characteristic functions identifies the increment law restricted to A as $\mathbb{P}(A)$ times the centered Gaussian law with covariance $(t-s)\text{Id}$. This proves the vector increment law and independence from the entire observation past. Repeating over successive time intervals gives all joint finite-dimensional Brownian laws. Continuous paths, starting at zero, and rational-time evaluations determine the law on the countable product of continuous-path spaces. This proves the claim. $\square$

4 A deterministic inverse and a Brownian sampler

The response estimate permits us to reconstruct the observations from their innovations. We first prove the deterministic statement on the full product path space. For a countable set I , write

$$\mathcal{C}_I = \prod_{i \in I} C([0, \infty), \mathbb{R}),$$

with the product of the compact-open topologies and its Borel sigma-field. For $z \in \mathcal{C}_I$, let $z(t) = (z_i(t))_{i \in I}$.

Lemma 4.1 (Deterministic inversion). *Let $b : [0, \infty) \times \mathbb{R}^I \rightarrow [0, 1]^I$ be jointly Borel. Suppose that a constant $0 \leq L < \infty$ satisfies*

$$\sup_{i \in I} |b_i(t, y) - b_i(t, y')| \leq L \sup_{j \in I} |y_j - y'_j| \quad (4.1)$$

for every $t \geq 0$ and every $y, y' \in \mathbb{R}^I$ whose coordinate differences have finite supremum. Then, for every $w \in \mathcal{C}_I$, there is a unique $Z(w) \in \mathcal{C}_I$ satisfying

$$Z_i(w)(t) = w_i(t) + \int_0^t b_i(s, Z(w)(s)) ds, \quad i \in I, \quad t \geq 0. \quad (4.2)$$

The map $Z : \mathcal{C}_I \rightarrow \mathcal{C}_I$ is Borel and causal: its restriction to $[0, T]$ depends only on the input restricted to $[0, T]$. It commutes with every coordinate relabeling under which the drift is equivariant.

Proof. All integrals below are ordinary scalar Lebesgue integrals. Define

$$Z^0(w) = w, \quad Z_i^{k+1}(w)(t) = w_i(t) + \int_0^t b_i(s, Z^k(w)(s)) ds, \quad k \geq 0. \quad (4.3)$$

These iterates are defined for every input and have continuous coordinates. Indeed, a continuous coordinate path family gives a Borel map into $\mathbb{R}^I$, so the integrand is Borel and bounded; its indefinite integral is continuous. The initial increment is bounded by t , uniformly in i . Induction using (4.1) gives

$$\sup_{i \in I} |Z_i^{k+1}(w)(t) - Z_i^k(w)(t)| \leq \frac{L^k t^{k+1}}{(k+1)!}, \quad k \geq 0, \quad (4.4)$$

where $L^0 = 1$, including when $L = 0$. At the induction step the differences of the two input fields are bounded by the preceding estimate, so the hypothesis of (4.1) applies.

For every finite T , the bounds in (4.4), with t replaced by T , have finite sum. Thus the iterates converge to a limit $Z(w)$ uniformly over all coordinates and all $t \in [0, T]$ in the sense that

$$\sup_{i \in I} \sup_{0 \leq t \leq T} |Z_i(w)(t) - Z_i^k(w)(t)| \rightarrow 0.$$

Each coordinate of the limit is continuous. The Lipschitz estimate then gives uniform convergence of the corresponding drifts on $[0, T]$, allowing passage through each integral in (4.3). This proves (4.2). The uniform estimates concern differences from the same input: the input itself may be unbounded over I , and its coordinates may have no common modulus of continuity.

For any other solution z , boundedness of b gives $\sup_i |z_i(t) - w_i(t)| \leq t$. Repeating the same induction yields

$$\sup_{i \in I} |z_i(t) - Z_i^k(w)(t)| \leq \frac{L^k t^{k+1}}{(k+1)!}.$$

Letting $k \rightarrow \infty$ proves uniqueness for every input.

We next verify measurability on this entire path space. The evaluation map $(s, z) \mapsto z(s)$ from $[0, \infty) \times \mathcal{C}_I$ to $\mathbb{R}^I$ is continuous. If Z^k is Borel, the integrand $(s, w) \mapsto b_i(s, Z^k(w)(s))$ is jointly Borel. Parameterized integration of a bounded Borel function shows that its integral from 0 to t is Borel in (t, w) . Consequently every rational-time coordinate evaluation of Z^{k+1} is Borel. Such evaluations generate the Borel sigma-field of $\mathcal{C}_I$, proving inductively that every Z^k is Borel. Their limits at rational times prove that Z is Borel as well.

Finally, the iteration uses only times up to its time argument, which proves causality. A coordinate relabeling that intertwines the drifts also intertwines each step of (4.3) and hence its limit. This applies equally to a bijection between two different countable index sets with corresponding drifts. $\square$

Here is the sampling consequence in a form that isolates the role of the response estimate.

Proposition 4.2 (Sampling from independent Brownian paths). *Let ν be a probability law on $\{0, 1\}^I$, where I is countable, and for each i choose finite sets $S_n(i)$ increasing to I . Interpret the finite-tilt functions (2.7) and the limsup drift (2.9) with this law and these sets. Suppose that the finite-tilt functions $b_{i, S_n(i)}(t, \cdot)$ are L -Lipschitz in the supremum norm on their finite fields, with one constant $0 \leq L < \infty$ for all i, n and $t \geq 0$. Then the drift has the Borel solution map of Lemma 4.1, and the Borel map $\Psi : \mathcal{C}_I \rightarrow \{0, 1\}^I$ defined by*

$$\Psi_i(w) = \mathbf{1} \left\{ \limsup_{n \rightarrow \infty} \frac{Z_i(w)(n)}{n} > \frac{1}{2} \right\} \quad (4.5)$$

sends the law of independent standard Brownian paths to ν . The map Ψ inherits every coordinate relabeling equivariance of the drift.

Proof. Each finite tilt is a ratio of finite sums with a strictly positive denominator, and is therefore jointly Borel in time and the field. Its limsup is jointly Borel and takes values in $[0, 1]$. If $d = \sup_j |y_j - y'_j| < \infty$, the assumed response bound gives

$$|b_{i, S_n(i)}(t, y) - b_{i, S_n(i)}(t, y')| \leq Ld$$

for every i, n, t . Taking limsups in both directions proves (4.1). Lemma 4.1 therefore applies. Formula (4.5) is a Borel predicate on every input, including those on which the displayed slopes do not converge.

To identify its full output law, use the reference observation model: take $X \sim \nu$, independently take standard Brownian paths (B_i) , and put $Y_i(t) = tX_i + B_i(t)$. Lemma 3.1 identifies the drift with the posterior mean at each fixed time, and Lemma 3.2 shows that

$$\widehat{W}_i(t) = Y_i(t) - \int_0^t b_i(s, Y(s)) ds$$

has the law of independent standard Brownian paths. By this definition, the identity

$$Y_i(t) = \widehat{W}_i(t) + \int_0^t b_i(s, Y(s)) ds$$

holds for all i and all t on every realization of the continuous paths. It does not require the fixed-time posterior identities to hold on one event simultaneously at every time. Deterministic uniqueness in Lemma 4.1 consequently gives $Y = Z(\widehat{W})$ as path families. Since Z is Borel, it follows that independent Brownian input w produces the complete joint path law of Y under Z .

For each $i \in I$ and $\delta > 0$, the Gaussian tail bound gives

$$\mathbb{P}(|B_i(n)| > \delta n) \leq 2 \exp(-\delta^2 n/2), \quad n \geq 1.$$

These probabilities are summable. Borel–Cantelli and a countable intersection over i and positive rational δ give $B_i(n)/n \rightarrow 0$ for every i simultaneously, almost surely. Hence $Y_i(n)/n \rightarrow X_i$ simultaneously, and $\Psi(\widehat{W}) = X$ almost surely. This identifies the entire configuration law of $\Psi(w)$ as ν . Equivariance follows from Lemma 4.1 and the coordinatewise formula (4.5). $\square$

Figure 1 separates the reference coupling used in the proof from the sampler driven by fresh independent noise. The same Borel maps are applied in both rows; the target sample is needed only in the reference argument.

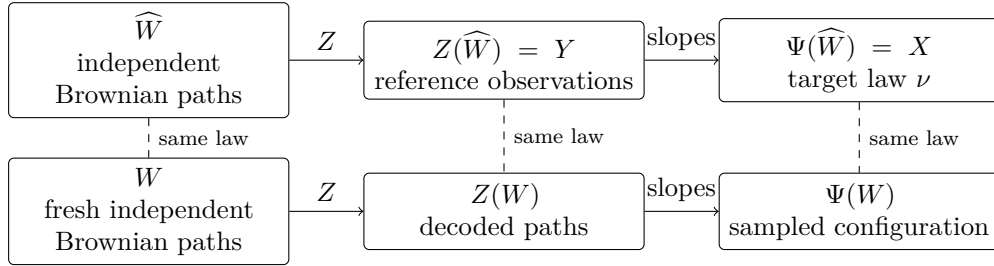


Figure 1: Law transfer in Proposition 4.2. The upper row is the reference coupling. Deterministic inversion identifies its middle term pathwise; integer-time slopes recover its right term almost surely. The lower row uses only independent input paths. Borel measurability transfers the full joint laws, not only the individual coordinate marginals.

Remark 4.3 (The uniformity hypothesis is substantive). Let I be countably infinite and let ν assign probability $1/2$ to each of the all-zero and all-one configurations. For nonempty finite S , all its finite-tilt means are

$$b_{i,S}(t, y) = \frac{\exp(a)}{1 + \exp(a)}, \quad a = \sum_{j \in S} (y_j - t/2).$$

At $a = 0$, the absolute row sum of its field derivatives is $|S|/4$. Thus no fixed L works along an exhaustion of I . The innovation lemma still applies to this law, but it alone gives no deterministic inverse: Proposition 4.2 uses the uniform response assumption precisely at that step. Likewise, equivariance of the sampler is asserted only for relabelings intertwining the chosen drift; arbitrary exhausting sets need not preserve every symmetry of ν .

For $\nu = \text{FUSF}_G$, the response bound (2.8) provides the hypothesis of Proposition 4.2 with $L = 1/2$. It remains to obtain the Brownian inputs from the vertex labels and to verify that the resulting construction is one Borel rule across graphs.

5 One Borel graph rule

We implement the preceding construction using only vertex labels and the distinguished undirected edge. Local finiteness and connectivity imply that each graph has countably many vertices and edges.

5.1 From vertex labels to independent edge paths

Lemma 5.1. *There is a Borel equivariant rule assigning a continuous path w_e starting at zero to each edge of a vertex-labeled graph such that, for every fixed graph with independent uniform vertex labels, $(w_e)_{e \in E(G)}$ are independent standard Brownian motions.*

Proof. Partition the binary digits of a uniform variable into countably many infinite subsequences. With a fixed Borel convention at dyadic rationals, this gives a Borel map from one uniform variable to a countable family of independent uniforms. At each vertex v , use this map to obtain a key K_v and a private sequence $(V_{v,r})_{r \geq 1}$, with all these variables independent. Almost surely the keys are distinct throughout the graph. On this event, an edge $e = \{u, v\}$ with $K_u < K_v$ selects $V_{u,r}$, where r is the rank of v among the neighbors of u , ordered by key. The rank is finite. Different edges select different pairs (u, r) : edges with different source vertices use different pools, and edges with the same source vertex have different other endpoints and hence different ranks. Simplicity is used here. Conditional on all the keys, the selected variables are distinct coordinates of an independent uniform family that is independent of those keys. They therefore have the product uniform law.

For completeness, one uniform can be mapped Borel measurably to a standard Brownian path. Split it into uniforms and apply the normal quantile to obtain independent standard normals, using finite defaults at exceptional quantile inputs. Start at $w(0) = 0$ and use some of them as independent increments on the nonnegative integer grid. On an interval $[a, a + h]$ of a dyadic grid, insert the midpoint value

$$\frac{w(a) + w(a + h)}{2} + \frac{\sqrt{h}}{2} \xi,$$

where ξ is a fresh independent standard normal. The two new increments are jointly Gaussian, have variance $h/2$, and have covariance zero. By induction, every dyadic grid has independent Gaussian increments of the appropriate variances.

On $[0, K]$, with K a positive integer, the refinement of mesh 2^{-p} uses $K2^p$ new normals. The probability that any has absolute value greater than p is at most $2K2^p e^{-p^2/2}$, which is summable for $p \geq 1$. The change of linear interpolation at that level is therefore eventually at most $p2^{-p/2}/2$. These bounds are summable. The interpolations converge uniformly on every compact interval almost surely. Their limit has continuous paths and the required dyadic Gaussian laws, hence the Wiener law by continuity. The event of compact-uniform convergence is Borel: it is the countable collection of uniform Cauchy conditions on integer intervals. Each condition can be tested at rational times for these continuous interpolations. Taking the limit on this event and the zero path otherwise defines a Borel path-valued map on every input.

Apply this same map to every selected edge uniform. This produces the required product Wiener law. On a labeling with any repeated keys, assign the zero path to every edge. A key collision is witnessed in some finite ball about the distinguished edge, so this exceptional case is Borel and equivariant. It has probability zero on each fixed countable graph. All other selections involve only finite neighbor lists and are Borel and equivariant as well. $\square$

5.2 Measurability on varying graphs

The decoder must be one rule on graph space, in addition to being Borel for each fixed graph. We give the details of this point. On the Borel set of labelings with distinct keys, vertices in any finite ball can be listed in increasing key order. Its undirected edges can then be listed lexicographically by $(\min\{K_u, K_v\}, \max\{K_u, K_v\})$. These are finite measurable listings, not a global ordering of the vertices. Re-distinguishing the graph at any edge selected from such a list is a Borel operation: each finite neighborhood of that edge can be read in a sufficiently large finite neighborhood of the

original edge. For example, if the selected edge lies in $H_r(e)$, its radius- q neighborhood is visible in $H_{r+q+1}(e)$. In particular, a Borel edge rule may be evaluated measurably at the finitely many edges in any such ball. The evaluated rule may itself depend on the entire graph; measurability follows by composition with re-distinguishing, not from finite dependence.

For the drift from Section 2, every $b_{e,S_n(e)}(t, y)$ is the limit of a finite weighted-tree ratio on $H_m(e)$ as $m \rightarrow \infty$, with n held fixed. The finite ratio uses only finitely many current fields and the finite graph. The outer limsup in n then defines $b_e(t, y)$. This exact order of limits makes the formula Borel when the current fields are supplied by any Borel edge rule jointly measurable in t .

Start with the paths $Z_e^0(t) = w_e(t)$ from Lemma 5.1. They are jointly Borel in the labeled graph, the distinguished edge, and t . If the same is true of Z^k , the preceding finite listings and drift formula show that

$$(G, U, e, s) \mapsto b_e(s, Z^k(s))$$

is Borel and bounded. Its scalar parameter integral over $[0, t]$ is Borel in (G, U, e, t) : on $0 \leq t \leq T$ it is the integral over $[0, T]$ of the bounded Borel function multiplied by $\mathbf{1}_{\{s \leq t\}}$. This is the usual measurability of integration with respect to a fixed finite measure, proved first for indicator rectangles and then by bounded approximation. Thus the next Picard iterate is jointly Borel. The countable limit defining Z , and the integer-time limsup in (4.5), preserve Borel measurability.

Every operation commutes with labeled isomorphisms. In particular, the balls $H_n(e)$ are centered on the unordered endpoints of e , and the finite key order is transported by any labeled isomorphism. Scalar time integration and the prescribed countable limits preserve this invariance. There is no distinguished vertex or external enumeration in the construction.

Proof of Theorem 1.1. On labelings with distinct keys, define $\Phi(G, U, e)$ by the readout (4.5), using the intrinsic forest drift and the edge paths of Lemma 5.1. On the remaining labelings set $\Phi = 0$ at every edge. The preceding argument makes this a single Borel, root-independent, equivariant rule, defined on all labelings.

Fix a graph G . Its key-collision event has probability zero, and the edge paths have product Wiener law. Lemma 2.3 gives the uniform finite-tilt response bound with $L = 1/2$ for $\nu = \text{FUSF}_G$. Proposition 4.2 therefore identifies the joint law of all the output indicators as FUSF_G . This proves the assertion for every fixed graph. $\square$

Proof of Corollary 1.2. Place independent uniform labels conditionally on (G, o) . The graphwise conclusion of Theorem 1.1, for the same Borel rule Φ , gives the desired conditional law. This is an equality of conditional probability kernels for each graph; it requires no intersection of label-conull sets over different graphs. Unimodularity imposes no further requirement on the construction. $\square$

6 Strongly Rayleigh processes on countable groups

The finite response hypothesis of Proposition 4.2 holds for every strongly Rayleigh law. The fixed-cardinality covariance identity for trees becomes an inequality for this larger class. The covariance-row estimate below appears in Alishahi, Barzegar, and Zamani [1, Lemma 2.5 and (1)], following the nonnegative-row-sum result of Ghosh, Liggett, and Pemantle [10, arXiv version, Lemma 3.2]. We give a proof by symmetric homogenization and apply it uniformly after positive external-field tilts.

Lemma 6.1 (Strongly Rayleigh response). *Let F be a nonempty finite set and let μ be a strongly Rayleigh probability law on $\{0, 1\}^F$. For $h \in \mathbb{R}^F$, let*

$$\mu_h(x) = \frac{\exp(\sum_{j \in F} h_j x_j) \mu(x)}{\sum_{z \in \{0, 1\}^F} \exp(\sum_{j \in F} h_j z_j) \mu(z)}, \quad p_i(h) = \mathbb{E}_{\mu_h} X_i.$$

For every $i, j \in F$, the derivative identity below holds, and for every $i \in F$ the row satisfies

$$\frac{\partial p_i}{\partial h_j} = \text{Cov}_{\mu_h}(X_i, X_j), \quad \sum_{j \in F} \left| \frac{\partial p_i}{\partial h_j} \right| \leq 2p_i(h)(1 - p_i(h)) \leq \frac{1}{2}. \quad (6.1)$$

Consequently

$$|p_i(h) - p_i(h')| \leq \frac{1}{2} \max_{j \in F} |h_j - h'_j| \quad (h, h' \in \mathbb{R}^F). \quad (6.2)$$

Proof. The tilted generating polynomial is

$$g_{\mu_h}(z) = \frac{g_{\mu}((e^{h_j} z_j)_{j \in F})}{g_{\mu}((e^{h_j})_{j \in F})}.$$

The denominator is positive. Positive coordinate scaling preserves stability, so μ_h is strongly Rayleigh; see Borcea, Brändén, and Liggett [4, Section 2.1(iv) and Proposition 3.1(2)]. Differentiating the finite partition sum gives the covariance identity in (6.1).

Write $n = |F|$ and $\eta = \mu_h$. To recover a constant total count, take a disjoint set F' of n dummy coordinates. First sample the occupied set $A \subseteq F$ with law η ; conditional on A , occupy a uniformly chosen subset of F' of size $n - |A|$. The resulting law $\tilde{\eta}$ keeps η as its F -marginal and has exactly n occupied coordinates in total. Write $\eta(A)$ for the probability of the occupied set A , and let e_k be the elementary symmetric polynomial of degree k in $w = (w_j)_{j \in F'}$. The generating polynomial of $\tilde{\eta}$, called the symmetric homogenization of g_{η} , is

$$\tilde{g}(z, w) = \sum_{A \subseteq F} \eta(A) \prod_{j \in A} z_j \frac{e_{n-|A|}(w)}{\binom{n}{|A|}}.$$

The polynomial has nonnegative coefficients, equals one at the all-one vector, is homogeneous of degree n , and specializes to g_{η} when all the w variables equal one. The additional fact that it is stable is supplied by [4, Definition 2.12 and Theorem 4.2]. Thus $\tilde{\eta}$ is strongly Rayleigh.

Let $C_{jk} = \text{Cov}_{\tilde{\eta}}(X_j, X_k)$. The Rayleigh inequality in [4, Theorem 4.1], evaluated at the all-one vector, gives $C_{jk} \leq 0$ for $j \neq k$. Since the total count is constant, for $i \in F$ we have

$$0 = \sum_{j \in F \sqcup F'} C_{ij} = C_{ii} + \sum_{j \in F \setminus \{i\}} C_{ij} + \sum_{j \in F'} C_{ij}.$$

The dummy covariances in the last sum are nonpositive. It follows that

$$- \sum_{j \in F \setminus \{i\}} C_{ij} = C_{ii} + \sum_{j \in F'} C_{ij} \leq C_{ii}, \quad \sum_{j \in F} |C_{ij}| \leq 2C_{ii}.$$

The covariances with both indices in F are those of η , and $C_{ii} = p_i(h)(1 - p_i(h)) \leq 1/4$. This proves (6.1), including degenerate coordinates. No homogeneity or full-support assumption on μ was used. Integrating the gradient along the segment from h to h' proves (6.2). $\square$

Proof of Theorem 1.3. Fix a site $i \in \Gamma$ and a finite observed set $S \subseteq \Gamma$. Let $J = S \cup \{i\}$ and let ν_J be the marginal of ν on J . For $t \geq 0$ and $y \in \mathbb{R}^\Gamma$, tilt ν_J by

$$\exp\left(\sum_{j \in S} (y_j - t/2)x_j\right).$$

The resulting probability law $\eta_{t,y}$ is an arbitrary positive external-field tilt of the strongly Rayleigh law ν_J , with zero log field on $J \setminus S$. Its mean at i is exactly the finite-tilt function $b_{i,S}(t, y)$ in (2.7). Finite differentiation and Lemma 6.1 give

$$\frac{\partial b_{i,S}}{\partial y_j}(t, y) = \text{Cov}_{\eta_{t,y}}(X_i, X_j) \quad (j \in S), \quad \sum_{j \in S} \left| \frac{\partial b_{i,S}}{\partial y_j}(t, y) \right| \leq \sum_{j \in J} |\text{Cov}_{\eta_{t,y}}(X_i, X_j)| \leq \frac{1}{2}.$$

Integrating along a segment of fields yields

$$|b_{i,S}(t, y) - b_{i,S}(t, y')| \leq \frac{1}{2} \max_{j \in S} |y_j - y'_j|. \quad (6.3)$$

For $S = \emptyset$, the finite-tilt function is constant and the same statement holds with the empty maximum defined to be zero. The bound is uniform in i, S, t and in all finite field values. It uses the actual finite marginal of ν , with no infinite external-field tilt or finite-volume approximation.

Choose finite sets $F_n \subseteq \Gamma$ increasing to Γ , with the identity in every F_n , and put

$$S_n(h) = hF_n, \quad h \in \Gamma.$$

These are finite exhaustions at every site, and $S_n(gh) = gS_n(h)$ for all $g, h \in \Gamma$. This requires neither finite generation nor amenability; for a finite group one may take $F_n = \Gamma$. Invariance of ν gives the exact finite-tilt identity

$$b_{gh,gS}(t, \tau_g y) = b_{h,S}(t, y) \quad (6.4)$$

for every g, h, t, y and finite S . Indeed, in the numerator and denominator of (2.7), substitute $x = \tau_g z$. Then $x_{gh} = z_h$, $(\tau_g y)_{gs} = y_s$, and $x_{gs} = z_s$, while the law of z is again ν . All sums are finite, so the identity holds pointwise even when the complete field is unbounded.

Define the limsup drift using these $S_n(h)$. Equations (6.4) and $S_n(gh) = gS_n(h)$ give

$$b_{gh}(t, \tau_g y) = \limsup_n b_{gh, S_n(gh)}(t, \tau_g y) = \limsup_n b_{h, S_n(h)}(t, y) = b_h(t, y).$$

Thus the drift is equivariant under every regular left translation on its entire domain. The uniform bound (6.3) supplies the hypothesis of Proposition 4.2 with $L = 1/2$. That proposition gives a Borel map Ψ from the product continuous-path space to $\{0, 1\}^\Gamma$, sending independent standard Brownian paths to ν and commuting with every τ_g on every input.

The proof of Lemma 5.1 constructs an everywhere-defined Borel map from one uniform variable to a standard Brownian path, with fixed defaults on its exceptional inputs. Applying this map coordinatewise sends independent uniform labels on Γ to independent Brownian paths and commutes with all translations. Its composition with Ψ is the required map Φ_ν . $\square$

The sets F_n above need not be Følner sets. Their translated form is available because the action is regular. For an arbitrary action on another countable set, equivariant finite exhaustions are not automatic, and no such extension is asserted here.

Proof of Corollary 1.4. For a countable discrete set, every self-adjoint positive contraction defines a determinantal probability law; see Lyons [11, Section 8]. For a finite $J \subseteq \Gamma$, let K_J be the compression of K to $\ell^2(J)$. It is a Hermitian positive contraction. For every $A \subseteq J$, the inclusion probability of A under the J -marginal is

$$\mathbb{P}(A \subseteq X) = \det K_{A,A} = \det(K_J)_{A,A}.$$

These inclusion probabilities determine the finite binary law by inclusion–exclusion. Hence that marginal is the determinantal law with kernel K_J , and it is strongly Rayleigh by [4, Proposition 3.5]. This covers complex Hermitian, singular, and projection kernels; no trace-class condition on the operator on $\ell^2(\Gamma)$ is needed. Theorem 1.3 now applies whenever the determinantal law is translation invariant.

If K commutes with the left regular representation, its matrix entries satisfy $K_{gh,gk} = K_{h,k}$. All finite inclusion determinants are then translation invariant. Inclusion–exclusion and the uniqueness of a countable law from its cylinder probabilities give invariance of the determinantal law, proving the final assertion. $\square$

References

- [1] Kasra Alishahi, Milad Barzegar, and Mohammadsadegh Zamani. On tail triviality of negatively dependent stochastic processes. *The Annals of Probability*, 51(4):1548–1558, 2023. Version cited: arXiv:2203.03935v2. <https://arxiv.org/abs/2203.03935v2>.
- [2] Omer Angel, Gourab Ray, and Yinon Spinka. Uniform even subgraphs and graphical representations of Ising as factors of i.i.d. *Electronic Journal of Probability*, 29:1–31, 2024. Paper No. 39. <https://doi.org/10.1214/24-EJP1082>.
- [3] Itai Benjamini, Russell Lyons, Yuval Peres, and Oded Schramm. Uniform spanning forests. *The Annals of Probability*, 29(1):1–65, 2001. <https://doi.org/10.1214/aop/1008956321>.
- [4] Julius Borcea, Petter Brändén, and Thomas M. Liggett. Negative dependence and the geometry of polynomials. *Journal of the American Mathematical Society*, 22(2):521–567, 2009. <https://doi.org/10.1090/S0894-0347-08-00618-8>.
- [5] Robert Burton and Robin Pemantle. Local characteristics, entropy and limit theorems for spanning trees and domino tilings via transfer-impedances. *The Annals of Probability*, 21(3):1329–1371, 1993. <https://arxiv.org/abs/math/0404048>.
- [6] Ahmed El Alaoui and Andrea Montanari. An information-theoretic view of stochastic localization. *IEEE Transactions on Information Theory*, 68(11):7423–7426, 2022. Preprint version: arXiv:2109.00709v2. <https://arxiv.org/abs/2109.00709v2>.
- [7] Ronen Eldan. Thin shell implies spectral gap up to polylog via a stochastic localization scheme. *Geometric and Functional Analysis*, 23(2):532–569, 2013. <https://doi.org/10.1007/s00039-013-0214-y>.
- [8] Ronen Eldan. Taming correlations through entropy-efficient measure decompositions with applications to mean-field approximation. *Probability Theory and Related Fields*, 176:737–755, 2020. <https://doi.org/10.1007/s00440-019-00924-2>.

- [9] M. Fujisaki, G. Kallianpur, and H. Kunita. Stochastic differential equations for the non linear filtering problem. *Osaka Journal of Mathematics*, 9(1):19–40, 1972. <https://ir.library.osaka-u.ac.jp/repo/ouka/all/5728/>.
- [10] Subhroshekhar Ghosh, Thomas M. Liggett, and Robin Pemantle. Multivariate CLT follows from strong Rayleigh property. In *2017 Proceedings of the Fourteenth Workshop on Analytic Algorithms and Combinatorics (ANALCO)*, pages 139–147. Society for Industrial and Applied Mathematics, 2017. Version cited: arXiv:1607.03036v1. <https://arxiv.org/abs/1607.03036v1>.
- [11] Russell Lyons. Determinantal probability measures. *Publications Mathématiques de l’IHÉS*, 98:167–212, 2003. <https://doi.org/10.1007/s10240-003-0016-0>.
- [12] Russell Lyons. ℓ^2 -Betti numbers, cost, and the free uniform spanning forest. *Oberwolfach Reports*, 10(3):2391–2392, 2013. Contribution to Report 42/2013, Group Theory, Measure, and Asymptotic Invariants. Published 1 June 2014. <https://doi.org/10.4171/OWR/2013/42>.
- [13] Russell Lyons and Andreas Thom. Invariant coupling of determinantal measures on sofic groups. *Ergodic Theory and Dynamical Systems*, 36(2):574–607, 2016. <https://doi.org/10.1017/etds.2014.70>.
- [14] Danny Nam, Allan Sly, and Lingfu Zhang. Ising model on trees and factors of IID. *Communications in Mathematical Physics*, 389(2):1009–1046, 2022. <https://arxiv.org/abs/2012.09484v2>.
- [15] OpenAI. The sharp factor-of-IID threshold for the free Ising model on regular trees. OpenAI Math Release preprint [OAI:The-sharp-factor-of-IID-threshold-for-the-free-Ising-model-on-regular-trees-September-26-2026](https://arxiv.org/abs/2012.09484v2), 2026.
- [16] Robin Pemantle. Choosing a spanning tree for the integer lattice uniformly. *The Annals of Probability*, 19:1559–1574, 1991. <https://arxiv.org/abs/math/0404043>.
- [17] Ádám Timár. Factor of iid’s through stochastic domination. *Israel Journal of Mathematics*, 2025. Published online 11 December 2025. Version cited: arXiv:2306.15120v2, 16 December 2025; first posted in 2023. <https://arxiv.org/abs/2306.15120v2>.